

Clarivi MVP

Scoping the smallest version that proves the idea | Apple Health Analytics Tool | Sabine Joseph

How to Use This Document

This is a thinking aid, not a decision. Sections 2 to 4 give a proposed starting position for the MVP boundary, written so you can accept, cut or move each item. Section 5 lists what should be investigated before building. Section 6 lists the decisions only you can make, and Section 7 the questions whose answers would change the scope. Nothing here is final until you say so.

1. The Core Value the MVP Must Prove

Core value in one sentence: Apple Watch records the numbers; this tool explains them. Using only their own Apple Watch data, a person sees in plain words what last night's readings mean against their own normal, why today's status was set, and one action for today, which Apple's own features do not explain.

An MVP earns its place by testing the three hypotheses this value rests on:

- **Feasibility:** data can flow from the watch to the tool reliably enough to support a daily signal.
- **Signal:** baseline deviation detection flags the days that actually felt off and stays quiet on normal days, often enough to be trusted.
- **Value:** the user reads the card and the nudge changes a decision at least some of the time.

Anything that is not needed to test one of these three is a candidate for cutting from the MVP.

Success criteria (D9): what makes the MVP done

A four-week tester period with all four users, judged as a mix: hard gates for Feasibility, target ranges plus end-of-test conversations for Signal and Value. The targets below are confirmed (29 September 2026) and must not change once results start arriving.

- **Gate, sync:** each user has a successful morning sync on at least 26 of 28 days.
- **Gate, push:** a notification is delivered on at least 90% of synced days that have a status (D36).
- **Gate, isolation:** no user ever sees another user's data, and the cross-user access test passes. Absolute, no tolerance.
- **Gate, correctness:** the SQL matches the pandas reference on your backfilled year, stage by stage, and no unfinished night is ever scored.
- **Signal target, your year:** of the disrupted days in your backfilled year (see below), at least 2 in 3 would have been called ease off or rest. No one is asked to remember how they felt; the check uses only evidence already in the data or in your calendar.
- **Evidence for your year:** a list of objectively disrupted days, built before testers start: travel days, found automatically from timezone changes in the samples; illness and major events, entered from your calendar into the event table; and workouts far below your usual duration or intensity for that activity. It only catches off days that left a trace, so a day that felt bad but changed nothing is missed; the daily check-ins during the test cover that gap.
- **Signal target, quiet by default:** ease off and rest together fire on no more than about 1 day in 7, on your year and during the test.
- **Signal target, testers:** the daily check-in ("off") and the tool's status agree more often than chance, reported per user over days that have a status (D36); with four people this shows direction, not proof. Check-ins can be skipped (D33), so the completion rate per tester is reported alongside.
- **Value target, engagement:** each tester taps the notification or opens the card on at least 5 days a week.
- **Value target, action:** on days the nudge says anything other than train as planned, testers mark it followed about half the time or more.
- **Value target, retention:** all three testers are still syncing and opening the card in week four.
- **Conversations:** a short conversation with each tester at the end: did the tool change a decision, what did they ignore and why, would they keep using it. Their answers carry as much weight as the percentages.

2. Proposed Must-Haves (the thin slice)

The smallest end-to-end path: ingest, clean, compute, show, nudge. Each item states why it is in, so you can challenge the reasoning rather than the list.

- **Data ingestion via automated sync:** each user installs a Shortcut on their iPhone that reads their Health data and posts it to an authenticated upload endpoint on the site. Run once, it backfills the last year of history in monthly chunks (Decision D26); run each morning by two automations (Decision D16): one when the phone is unplugged from its charger, and a catch-up when the tester first opens an everyday app of their choice, it posts the previous night and day. There is no manual export in the MVP (Decision D2). Each post is JSON carrying raw samples, not daily summaries (Decision D18): every reading with its timestamp, timezone offset, source device and unit, plus a schema version, a backfill flag and a month ID; the user is identified by their upload token, never by the body. Resting HR, HRV, sleep stages and respiratory rate are sent in full; heart rate is sent only for a fixed, generous overnight window (6pm to noon), which the server trims to the actual sleep window. The endpoint is idempotent, so a Shortcut that runs twice or is re-run by hand never duplicates data. For your account only, the backfill also fetches workout summaries (activity type, date, duration and average heart rate), used solely to build the disrupted-day evidence for the Signal check (D9). Wrist temperature is not collected in the MVP. The source stays pluggable, so a HealthKit app or bridge app can replace Shortcuts later without touching anything downstream.
- **Two morning triggers:** the same Shortcut runs from two personal automations (Decision D16). The wake trigger fires when the phone is unplugged from its charger, which for anyone who charges overnight is a reliable sign they are up; it needs no iPhone alarm. Unplugging does not unlock the phone, so this run may still be blocked. The catch-up trigger fires when the tester opens an everyday app they choose during onboarding, which guarantees the phone is unlocked. Before doing anything, the Shortcut checks the time (it acts only between 4am and noon) and whether today's sync already succeeded, and stops if so; a midday or evening unplug therefore does nothing. The server then decides whether the night is actually over (R24), so whichever run first finds a finished night produces the score and the morning push; any later run changes nothing.
- **Per-user upload tokens:** each user's Shortcut authenticates with its own upload token, generated in the app. A token can write that user's data and nothing else, can never read, and can be revoked from the app. Because the token travels inside the configured Shortcut, a configured Shortcut must never be shared (Risk R19).
- **Cleaning and a daily feature table:** explicit day-boundary and timezone rules, coverage per day, and three simplifying rules that keep the cleaning in plain SQL (Decision D10). One source: every metric is taken from the Apple Watch only, since every user wears it overnight and the iPhone does not measure heart rate, HRV or respiratory rate, which removes most overlapping-source handling. One sleep window: each night's window runs from the first to the last "asleep" sample in the Watch's own sleep stages, and overnight heart rate is trimmed to it with a simple join. One definition: sleeping HR is the median heart rate during asleep periods. Because the Shortcut sends raw samples (D18), all of this happens on the server; the database keeps the raw samples and builds the clean daily table from them, so a rule change is a recompute, never a re-sync. Any step that still proves awkward in SQL moves to a Supabase Edge Function in the same project.
- **Minimal data quality gate:** a per-day confidence label; low-coverage days are excluded from baselines and marked on the card. Trust in the signal depends on it.
- **Baseline drift and anomaly detection on three metrics:** sleeping HR, overnight HRV and sleep duration (Decision D5). Each metric is compared with the user's own baseline: the median of the previous 28 days, never including the day being judged, with a flag when a reading sits more than 2 scaled MADs from it. A scaled MAD is the median absolute deviation multiplied by 1.4826, so that it matches a standard deviation while still ignoring the odd extreme night. No flag is raised unless the window holds at least 21 valid days, so a stretch of missed nights suppresses flags rather than producing unreliable ones; while a baseline is still building, D36 decides what the card shows. The same fixed defaults apply to everyone (Decision D11). With scaled MAD, per-metric flags are uncommon (roughly 1 day in 45 per metric in one direction, if readings are bell-shaped), so a flag means something; the readiness thresholds (D24) still decide what the nudge says. Written as SQL (window functions and percentile_cont) and run each time new data arrives for a user (Decision D19), filling the insights table. Note that D5 scopes the score, not what is ingested: the illness composite below needs two extra daily values that the Shortcut can carry at almost no cost.
- **Readiness score:** rule-based with fixed weights (Decision D6): each of the three metrics is expressed as a deviation from the user's own 28-day baseline, the deviations are combined with fixed weights, and two thresholds separate ready, ease off and rest. Fully expressible in SQL and fully explainable: the card can show the arithmetic. The initial weights and thresholds, and whether the card shows how the status was decided (the Why today screen) or just the result, are Decision

D24. Parameters are stored, not hard-coded, so user-tunable or learned weights can follow later without a rewrite. A night with a reading missing is scored from the other two (D35).

- **Today's readiness card:** status (ready, ease off, rest), the two or three metrics driving it in plain language, the nudge with its one-sentence reason, and the data freshness, laid out to fill the first screen of an iPhone (Decision D27). The full arithmetic sits one tap away on the Why today screen (Decision D24), titled with the day's status, for example "Why ease off today". Each reading is shown against the tester's normal range in plain words, with its numbers (normal range, difference from normal, and how it compares with the last 4 weeks) on tap. A closing section explains how the status is decided, with each reading's points, the day's total and the ease off and rest thresholds on tap; the weights themselves are not shown, only the order in which the readings matter. Opens of that screen are logged, so you learn whether testers check how the status was decided.
- **Nudge:** one action from a fixed vocabulary (train as planned, train easy, rest, prioritise sleep). Sent each morning as a push notification to the tester's iPhone and shown on the card (Decision D4); tapping the notification opens the card in the home-screen app. The notification carries the status and the reason, for example "Ease off today: HRV well below your usual, sleep short" (Decision D29). Computed as soon as the morning sync lands (Decision D19) and stating its freshness plainly ("as of this morning's sync"). Daily cadence is the target (Decision D12), which makes the sync spike in Section 5 the gating item (A7).
- **Minimal trend view:** for each metric in the score, one line with the baseline band and flagged points, sized for a narrow iPhone screen first (D27). This is the detailed dashboard reduced to the minimum that lets a user verify the card.
- **Accounts, login and data isolation:** each user signs in with an email address and password (Decision D30), using Supabase's built-in login, and sees only their own data. Sign-up is closed to the public: you create each tester's account yourself in the Supabase dashboard, so nobody else can create one, and the MVP sends no email at all (no invitations, confirmations or reset emails). Each tester gets a temporary password from you privately and changes it once in the app. On the iPhone, iCloud Keychain saves the password and fills it with Face ID, so day to day it feels close to Face ID sign-in. Row-level isolation between users is a correctness requirement, not a feature: one user seeing another's health data is the worst failure this MVP can have (Risk R13). Because testers use the tool from an iPhone every morning, sessions are long-lived, balanced by a sign-out-everywhere control and re-entering the password before sensitive actions (reissuing an upload token, deleting data); a 30-day idle expiry follows once the project moves to Pro after the test, since it needs the paid plan (D37). Includes an in-app change-password screen and a minimum password length; a forgotten password is reset by you, setting a new temporary password and telling the tester privately. Nothing fancier.
- **Installable iPhone web app:** the site is mobile-first and installable to the iPhone home screen as a web app (Decision D27): an app icon, full-screen display, and no App Store. Laptop use still works but is the adapted layout, not the primary one: the same single column as on the iPhone, centred, at most 480 points wide (D39).
- **Push notifications:** iOS delivers web push only to web apps added to the home screen, and only after the user grants permission from a tap inside the app, so the onboarding flow asks for it once, after install. The server stores one push subscription per device, sends the nudge once the analysis triggered by that user's upload has finished (D19), and logs whether each notification was delivered and tapped. At most one morning nudge per user per day: if the Shortcut runs twice, or a tester re-runs it by hand, the second analysis updates the card but does not send a second nudge. On days the nudge asked for a change (train easy, rest, prioritise sleep), an evening follow-up push goes out at 8pm local time asking whether they followed it (Decision D32); it is not sent on train-as-planned days or when no morning nudge went out. Its wording carries no health detail, since it also shows on the lock screen (R23). The server works out each tester's local 8pm from the timezone offsets in their latest samples. If no sync has succeeded by 11:30am local time (Decision D34), a reminder push goes out: "No sync yet this morning. Run your readiness Shortcut before noon to get today's nudge." It carries no health detail and is sent at most once that day. A sync that lands between 11:30am and noon, whether by hand or by a late automation, still produces today's card and nudge, marked as late, and the 8pm follow-up still applies on change days. Only if nothing syncs by noon does the day go without a nudge (R1), because the Shortcut's window stops runs after noon.
- **Analysis on upload:** each accepted upload triggers that user's analysis (baselines, score, composite, nudge) for the affected dates, and then the push (Decision D19). The weekly digest remains a scheduled job. Backfill chunks must not each trigger a full recompute and a notification: backfill uploads are marked as such, recomputed once when the backfill completes, and never send a push.
- **Usage and outcome logging:** server-side record of logins, card views, flags shown, nudges sent and tapped, and whether the user marked the nudge followed, including when and from where (the evening push, the card after 8pm, or the next morning's card). Because everything runs on your server, this replaces any returned file and is your direct evidence for the Signal and Value hypotheses. It settles D8: the feedback button is in.

- **Evening follow-up:** on days the nudge asked for a change, the question "Did you follow today's nudge?" appears only from 8pm local time (D32). At 8pm a push asks it, and from then the card shows it with yes and no above the morning's briefing; tapping the push opens the card on the answer. Before 8pm the card shows the nudge without the question. The answer can be changed until the next morning. If the tester still has not answered by the next morning, that morning's card asks about yesterday before showing today's status. Train-as-planned days are never asked, which matches the D9 action target.
- **Daily check-in:** one tap on the card each day: how do you feel today, good, okay or off. It is shown before the status is revealed when the card opens, stored alongside the day's score, and never feeds the score, so it stays an independent measure of how the day actually felt (D9). Testers can skip it (D33): the card then appears straight away and keeps a small prompt so they can answer later that day. An answer can be changed until the next morning's check-in appears. Every answer is logged with its time and whether the tester had already seen the status, and the Signal check uses the first answer given before the status was seen wherever one exists. Because the notification already shows the status and reason on the lock screen (D29), a tester may see the verdict before answering; that bias is noted in the results rather than designed away.
- **Consent and security:** written consent from each tester; HTTPS everywhere; encryption at rest for the database and all synced data; per-user upload tokens that can write but never read (R19); access to production limited to you; a retention rule and deletion on request; no third-party analytics scripts on pages that show health data. You are now holding other people's health data on a server you operate, so this is a must-have (Risk R8, back at full strength).
- **Tester onboarding:** a one-page guide covering adding the web app to the home screen, signing in from inside that home-screen app (not from Safari) with the temporary password you give them, changing it in the app and saving the new one to iCloud Keychain, allowing notifications, installing the Shortcut and adding their upload token, running the one-time backfill, setting up the two morning automations and choosing their catch-up app, and what to do if a morning run fails (run the Shortcut by hand), plus what the card means, how to mark a nudge followed, and contacting you directly if they forget their password.

Decided in (D8): three items that were borderline

- **Illness and strain composite:** in, confirmed. A multi-signal flag that fires only when several overnight markers move together in the same direction. The score stays at three metrics; the composite additionally reads Apple's resting HR (one daytime value Apple publishes at the end of each day, so a morning composite uses yesterday's) and respiratory rate where a tester's watch records them, and degrades gracefully when they are absent, stating which inputs it had. Wrist temperature is excluded from the MVP entirely and can be added to the composite later as one more optional input. This fixes the extra columns in the daily file (see Data ingestion). Brings R7 (health-adjacent wording) into the must-do list: the composite is a pattern note, never a diagnosis.
- **Weekly digest:** in. A plain-language summary each week of what moved against baseline, what was flagged, how many days had data and how many nudges were followed, shown on a page in the app. Written from a fixed template filled from the insights table for the MVP (Decision D22); a model-written version, constrained to the same numbers and checked against a list of forbidden claims, is a later feature the architecture should allow for. Shown as an in-app page only (Decision D23).
- **Nudge feedback capture:** decided in, as part of usage and outcome logging above.

3. Proposed Out of Scope for the MVP

Out of scope means not built for the MVP, not abandoned. Grouped by the reason for exclusion:

Needs more history than an MVP will have

- Lagged cross-metric relationships (60 to 90 days), personal sleep need estimation, seasonality and periodic decomposition, changepoint detection, and the causal layer (guided self-experiments and natural experiment analysis). Decided: the causal layer is out of the MVP and is presented as method and design, with its schemas kept in the architecture (Section 4, Decision D7).

Needs a workout-level data model the core signal does not require

- Strength session analysis, heart rate recovery, intensity distribution across cardio, standardized fitness test tracking.

Valuable but not on the core value path

- Weight and body composition trend, mobility and gait metrics, sedentary patterns, behavioural and adherence patterns, nudge calibration loop, the full progressive-disclosure dashboard, and the event annotation user interface (the schema stays in, see Section 4).

4. Future Features the Architecture Should Support Now

These are cheap to design in and expensive to retrofit. None of them add MVP features; they keep the door open.

- **Keep raw samples, not just daily aggregates:** the MVP stores every sample the Shortcut sends (D18) and builds the daily table from it. Heart rate recovery, strength analysis and intensity distribution will also need daytime heart rate, which the MVP does not fetch; adding them means widening the Shortcut's heart-rate window and extending the backfill, with nothing already stored redone.
- **A module contract:** every analysis module reads the daily feature table and emits insights in one standard shape (metric, date, value, baseline, deviation, severity, explanation) into an insights table. The card, digest and nudge only ever read that table, so new modules plug in without touching the interface. This also lets the analysis live wherever is convenient, a database function today or a Python service tomorrow (D10 option B), without the app noticing.
- **Event annotation schema:** a table of dated events with a type and a note. The MVP uses it straight away: you enter illness and major events from your calendar for your backfilled year, and travel days are added automatically, to form the disrupted-day evidence for the Signal check (D9). No interface is needed yet; entries can go in directly or from a small file. The causal and correlation modules will depend on the same table later.
- **Experiment registry schema:** hypothesis, outcome metric, baseline and test windows, status, timestamps. Pre-registration is only credible if the record exists before the result. With the causal layer out of the MVP, this schema is what lets the design be shown as ready to run.
- **Nudge log with feedback fields:** date, nudge, reason, followed (yes/no), next-day outcome. Enables calibration later and doubles as evidence for the Value hypothesis.
- **Workout and segment tables:** workout id, type, start and end, lap markers, linked HR series. Populated by the parser now, consumed later.
- **Delivery channel abstraction:** card, digest and nudge as content objects with pluggable channels (in-app, web push, email), so changing how things arrive does not mean rewriting them. The MVP uses in-app and web push for the nudge (D4) and in-app only for the digest (D23).
- **Digest writer abstraction:** the digest job separates computing the week's facts from writing the sentences, so the template writer can be replaced by a guarded model-written one (D22) without touching the numbers.
- **Sync-path abstraction:** ingestion as a pluggable source (a Shortcuts automation now; a HealthKit app or bridge app later) with the same parsed output, so switching sync path is a new adapter, not a rewrite.
- **User scoping done properly:** multi-user is now a must-have (Section 2), so this is about doing it well: user id on every table, isolation enforced in one place (a query layer or row-level security), and an automated test that tries to read another user's data and fails.
- **Deployment discipline:** a single deployed version means no version drift across testers, but changes now affect everyone at once. Keep a changelog, tag releases, and freeze score logic during a test period so evidence stays comparable. Every change first goes to a Vercel preview address (D21) and is checked on your own phone before it goes live.
- **Configurable parameters:** baseline windows, thresholds and score weights stored as parameters, not constants.

5. Investigate Before Building

Product assumptions to test

- **A1** Users will act on a daily nudge. You can test this on yourself for two to three weeks with a hand-computed nudge from your own data, before any interface exists.
- **A2** A readiness signal is trusted without external validation. Check flagged days in your backfilled year against objectively disrupted days (travel, calendar illness and events, and workouts far below your usual), not against memory; during the test, the daily check-ins provide the in-the-moment comparison.
- **A3** Target users wear the watch overnight. HRV, sleeping HR and respiratory rate all depend on it. Confirmed for you and all three testers.
- **A4** Retired for the MVP: wrist temperature is not collected for the score or the composite. It remains relevant only if a later version adds it as an optional composite input (Series 8, Ultra and later only).

- **A5 Retired:** no second app is needed. Testers' iPhones run the Shortcuts automation for both the backfill and the daily sync (D2), which they have confirmed they can do.
- **A6** At least 28 days of history exists before the tool is useful. Met for you with five years of history; confirm each tester has at least 28 days, which the backfill then covers on day one.
- **A7** The Shortcuts automation runs reliably every day on testers' phones without them doing anything. Unproven: iOS protects Health data while the phone is locked, so locked-phone behaviour and background limits (R9) are the main unknowns. Test on your phone and one tester's in week one, with both triggers (D16).
- **A11** Every user charges their iPhone overnight, so unplugging it in the morning marks waking. Confirm this at onboarding. For anyone who charges during the day instead, the morning unplug never happens and the charger trigger becomes just a second catch-up; for them, swap the wake trigger for Sleep Focus turning off or a fixed early time, with the catch-up trigger unchanged.
- **A8** Resolved: you wear an Apple Watch Ultra (1st generation), and testers wear Ultra (1st generation) and SE (2nd generation), all overnight; all five MVP metrics are recorded on both models. Neither model can run watchOS 27, so Apple's new Readiness score, Recovery HRV and five-second heart rate (Series 12 and Ultra 4 only) are on no user's wrist, and your backfilled year is in the same data format throughout. Every user, you included, keeps the same watch through the test, since a new model would restart that person's baselines. The SE 2's lack of wrist temperature no longer matters now that temperature is out of the MVP. Absolute HRV readings can differ between models; that is harmless because every baseline is personal, but testers should never be compared to each other on raw values. The score must still cope with an occasional missing night.
- **A9** Testers will sign in to the account you create for them, install a Shortcut on their iPhone, and log in daily from any device. The automation posts to your upload endpoint each morning; if a run fails they can run the same Shortcut by hand. Confirmed: testers have iPhones that run automations and have agreed to server-side storage. Daily login remains to be observed.
- **A10** SQL can carry the MVP analysis. Baseline median and MAD, rolling windows and a weighted score are expressible in Postgres, and the D10 simplifying rules (Watch-only sources, Watch sleep stages as the sleep window, sleeping HR as a median) keep the cleaning within plain SQL, with Supabase Edge Functions for anything that is not. With the Shortcut posting raw samples as JSON (D2, D18), no export parsing is needed at all. Later modules such as seasonality decomposition or the causal layer will likely exceed SQL and be the point at which a Python service is added through the module contract.

The chosen sync path: iPhone Shortcuts (spike first)

The iOS Shortcuts app has a "Find Health Samples" action that can pull readings for selected types over a date range, and a "Get Contents of URL" action that can post them to a web address as JSON. The same Shortcut does two jobs: run once, it backfills the last year of history, month by month (D26); run by two morning automations (the phone being unplugged from its charger, and a catch-up when an everyday app is opened; D16), it posts the previous night and day. The unknowns to settle in a half-day spike on your phone and one tester's: whether a scheduled run can read Health data when the phone is locked (iOS protects it while locked); how often the unplug run finds the phone locked and has to wait for the catch-up, and how long after waking the nudge typically arrives; how long a one-year backfill takes and whether it completes reliably (R20); and how large a month of raw samples is in practice, dominated by overnight heart rate (D18). A rough estimate is a couple of hundred heart-rate readings a night, about 70,000 a year, or a few hundred kilobytes per monthly post. If the spike fails, the upgrade path is a HealthKit companion app, not a return to manual exports.

Dependencies

- **Sync path:** an iPhone Shortcut on each user's phone (D2). No app to build or install; depends on Apple's Shortcuts Health actions and automation behaviour, which can change with iOS updates. A HealthKit companion app remains the upgrade path if Shortcuts proves unreliable (Xcode, Swift, an Apple Developer Program membership at \$99 a year, and TestFlight).
- **Email delivery:** none in the MVP (D30). You create each tester's account in the Supabase dashboard and reset forgotten passwords yourself. Supabase's built-in email sender only reaches members of your own Supabase team, at 2 messages an hour, so it could not send tester invitations anyway; a wider rollout would need an email provider connected to Supabase and a domain you own.
- **Hosting and operations:** Vercel serves the front end (the React code, icon and service worker) from your repository, and Supabase hosts the database, login, storage, upload endpoint and push sending (D17, D21). Health data never passes through Vercel: the phone talks to Supabase directly. Both stay on free plans through the build and the four-week test, and Supabase moves to Pro after the test (D37). On Supabase, encryption at rest is on every plan; the free plan has no

automatic backups, so you make your own with the Supabase CLI's db dump command, saved to an encrypted folder on your laptop at least weekly and before every database change (R8). There is one Supabase project, with a local copy on your laptop for building and testing database changes (D38). Vercel's free plan covers personal, non-commercial use, which fits a capstone. Someone (you) is on the hook when either is down.

- **Notification channel:** web push to the home-screen web app (D4, D27). Needs a key pair for signing notifications (VAPID), a table of push subscriptions, and a small sending function on Supabase, all standard for web push. No third-party notification service is required.
- **Test data:** your five years of history plus three testers' histories, with consent to server-side storage confirmed (Risk R8).
- **Class timeline:** no fixed due date for the MVP or the final demo, and about 40 hours a week available, and no self-set milestone (D25). Time is not the constraint; scope is. With neither a date nor a milestone, the success criteria (D9) define when the MVP is done: a four-week tester period judged against the gates and targets in Section 1.

Technical risks

Risk	Why it matters	How to de-risk
R1 Stale data when a sync is missed	A missed or failed automation run means the nudge is computed on old data, which can be wrong for today, and a wrong nudge costs more trust than no nudge.	Show freshness on the card and each user's last sync time. If no sync has succeeded by 11:30am local time (D34), send one reminder push and show the missed sync on the card; a sync before noon still produces today's nudge, marked as late; with no sync by noon, the day gets no nudge. Reminders on consecutive days point to a broken automation, so flag them to yourself.
R2 False positives from periodic patterns	Weekly schedule and monthly rhythms shift sleeping HR and HRV on a predictable cycle. Without decomposition the detector fires on them, causing nudge fatigue and lost trust.	Quiet by default: scaled-MAD metric flags (D11) are already uncommon, and the readiness thresholds (D24) should keep ease off and rest uncommon too. Measure on your backfilled year how often each flag and each nudge would have fired. Consider a simple day-of-week adjustment in the MVP if weekly patterns dominate the flags.
R3 Timezone and day-boundary errors	Travel days produce corrupted daily aggregates and phantom anomalies.	Write the day-assignment rule explicitly (sleep to wake date, local time from record offsets) and unit-test it against a known trip.
R4 Export size and parse time (retired unless the fallback is used)	No export file is parsed in the MVP (D2). Applies only if the Shortcuts backfill fails and a one-time export plus converter script is needed as a fallback; your five-year export would likely exceed a gigabyte of XML.	If the fallback is triggered: a streaming parser (iterparse) in a local converter script that writes the same JSON sample format the Shortcut posts (D18).
R5 HRV sparsity and missing nights	SDNN readings are few and mostly overnight; a night off the wrist leaves the score with a missing input.	Score degrades gracefully (D35): a missing reading's weight moves to the other two and the card says it used 2 of 3 readings; with two or more missing, or no sleep recorded, the day gets no status.
R6 Duplicate and overlapping sources	When more than one device writes the same metric, naive processing double counts or mixes readings of different quality.	Watch only as the source for every MVP metric (D10), which sidesteps most overlap; log and ignore readings from other sources; if daytime metrics such as steps are added later, use interval-union summing tested against Apple's own daily totals.
R7 Health-adjacent claims	"Illness warning" and "rest today" carry responsibility if wrong or over-read.	Define the language and a clear not-medical-advice position before the first user sees a flag.

Risk	Why it matters	How to de-risk
R8 Privacy and security of health data	You hold other people's health data on a server you operate. Testers have confirmed they are comfortable with this and will sign consent, which settles consent but not the rest: HTTPS, encryption at rest, access control, retention and deletion all apply, and a breach would be serious for the testers and for you.	Now a must-have (Section 2). Keep the tester group small, store only what the modules need, and state the security posture plainly in the capstone. Check whether your employer or local data-protection rules have anything to say about a personal project holding third-party health data.
R9 Shortcuts reliability: locked phone and background limits	iOS protects Health data while the phone is locked, so a scheduled run may fail or return nothing if the phone is locked at that moment. Automations can also be delayed or skipped. Each silent failure makes the nudge stale.	Two triggers (D16): the wake trigger may fire while the phone is still locked, and the catch-up trigger, which fires when an app is opened, guarantees an unlocked run. Measure at the spike how often the catch-up has to do the work. Keep the endpoint idempotent, the Shortcut's 4am-to-noon window and "already synced" check, last sync time on the card, and a flag for any user with no successful sync that day. If both triggers fail often, move to a HealthKit app.
R10 Nudges that are sent but not seen	A pushed nudge helps only if the tester sees and acts on it. A tester who declines notifications, silences them, or swipes them away gets no nudge, and the Value hypothesis cannot be tested on them.	Log notifications delivered and tapped, and card views, per tester per day; the engagement target in D9 is at least 5 days a week. Push (D4) is the main mitigation; long-lived sessions (D27) stop a login screen from getting in the way when a notification is tapped.
R11 Upload and onboarding failures	A tester who cannot install the Shortcut, add their token, or keep the automation running produces no data, and the failure is invisible to you until you notice the gap.	A dry-run week with one tester; a status page per user showing last successful sync; running the Shortcut by hand as the fallback; a support channel.
R12 Deployment breaks everyone at once	A single hosted version means a bad release or a schema change affects every tester simultaneously and mid-test.	Check every change on a Vercel preview address (D21) on your own phone before it goes live; freeze score logic during test periods; tag releases and keep a changelog. With one Supabase project (D38), previews talk to the live database: test every database change on the local copy first, apply changes only at release and in a backward-compatible way, check previews signed in as yourself or a test account, never as a tester, and back up before every change during the test.
R13 Data isolation bugs	A query missing a user filter shows one tester another tester's health data. This is the single worst thing the MVP can do.	Enforce isolation in one place (database row-level security), never in individual page code, and write a test that attempts cross-user access and must fail. Uploads are the exception: the Shortcut posts with its upload token, not a login, so the upload function writes with Supabase's service key, which bypasses row-level security. That function looks up the user from the token on the server and never takes a user id from the request, and the cross-user test covers this path too: one tester's token cannot write rows for anyone else, and a token can never read data back.

Risk	Why it matters	How to de-risk
R14 Hosting cost and operational load	Even a small stack costs money monthly and needs someone to patch, back up and restart it. With a builder and Supabase most of this is bundled. With the free plan through the test (D37), the stack costs nothing until after the test, but there are no automatic backups and the project can pause after a week of low activity; Supabase's paid tier adds daily backups and stops pausing.	Make your own db dump backups at least weekly and before every database change, practise one restore before the test, and set a calendar reminder for backups and dependency updates; daily syncs keep the project active; move to Pro after the test (D37); automate deploys.
R15 Low-code limits on heavy data work	Largely retired by D2: no export is parsed, because the Shortcut posts raw samples as JSON (D18). What remains: Shortcuts' own limits on run time and payload size for the backfill (R20), and the risk that generated code leaves tables without row-level security switched on.	Keep the posted data compact and chunk the backfill by month. Verify row-level security is enabled on every table by testing with a second account.
R16 Generated SQL statistics that are subtly wrong	AI-generated SQL is prone to quiet errors: off-by-one windows, mixing UTC and local dates, medians over the wrong partition, and sleep-window trimming or sleeping-HR derivation that looks right but drops or double-counts readings. The simplifying rules in D10 shrink this surface, but a wrong baseline still produces confident wrong nudges.	Keep a small local pandas notebook that pulls your own raw samples from Supabase and recomputes both the cleaning (daily values, sleep windows, sleeping HR) and the baselines as a reference implementation. Run both over your backfilled year and require the SQL to match, stage by stage, before any tester sees a flag; rerun the comparison whenever the SQL changes.
R17 Owning generated code (revised by D20)	With Claude Code the code is yours, which removes builder lock-in but means you own a repository, a deploy pipeline and any bugs the generator leaves behind. Low-code in the sense of not writing it, not in the sense of not running it.	Keep the project small and conventional (React, Supabase, one schema); ask Claude Code for tests alongside features, especially the cross-user access test (R13) and the SQL-versus-pandas check (R16); commit often so any step can be undone.
R18 Scope creep without a deadline	With no due date and 40 hours a week, the easiest failure is adding modules (lagged correlations, strength analysis, the causal layer) before the thin slice has proven the core value. Each addition delays the only evidence that matters: testers using the card.	D9 is set, so hold to it: the MVP is done when the four-week tester period has run and been judged against the gates and targets in Section 1. Treat Section 3 as a parking lot, not a queue; add nothing new until then.
R19 Upload tokens inside a Shortcut	The Shortcut has to authenticate without a browser login, so a token lives inside each user's configured Shortcut. If that Shortcut is shared or the phone is compromised, someone else could post data as that user.	Per-user, write-only, revocable tokens generated in the app; a visible revoke-and-reissue button; never distribute a configured Shortcut, only the blank template; log every upload with its token.
R20 Deep backfill through Shortcuts	Heart rate is the heavy part of a backfill, since sleeping HR derives from every overnight reading. Even with heart rate limited to an overnight window (D18), a year of raw samples is tens of thousands of rows per user, and a Shortcut may be slow, hit time or memory limits, or fail partway.	Backfill depth is set at one year (D26), fetched in twelve monthly chunks; make each chunk idempotent so a failed run resumes from the last completed month; time the backfill on your phone and a tester's at the spike; if chunks are too heavy, narrow the overnight heart-rate window before anything else (D18); keep the one-time export plus converter script as the documented fallback.

Risk	Why it matters	How to de-risk
R21 Home-screen web app quirks on iOS	On iOS a home-screen web app keeps its own storage, separate from Safari, so a login in one does not carry to the other; testers must sign in inside the home-screen app itself. iOS may also clear storage for web apps that are not used. With D28 withdrawn, the risk of a Shortcut opening the card in Safari no longer applies.	Onboarding has testers sign in from inside the home-screen app, after adding it, not from Safari; keep sessions long-lived; test at the app spike that tapping a notification opens the home-screen app still signed in; if storage is cleared, the app asks the tester to sign in again (iCloud Keychain fills the password) and re-registers for push.
R22 Web push reliability on iOS	Push only works for web apps on the home screen with permission granted. Deleting and re-adding the app, clearing website data, or a Focus mode can silently stop delivery, and subscriptions can expire. A missing nudge looks to the tester like the tool did nothing.	Log delivery per notification and flag users with no successful delivery in two days; re-register the subscription each time the app opens; show on the card whether notifications are on and when the last one was delivered. With D28 withdrawn there is no second route, so a delivery failure should be visible to you the same day.
R23 Health details on the lock screen	Notification text appears on the lock screen and in notification history, where anyone near the phone can read it. A message such as "rest today: HRV well below your usual" shares health information the tester may not want visible.	Decided: status and reason (D29), accepted for its usefulness. Mitigate rather than avoid: say plainly in the consent that nudges with their reason appear on the lock screen; show testers how to set this web app's notification previews to "When Unlocked" in iOS settings if they want privacy; apply the R7 wording rule to notification text; never name a condition or suggest illness in the notification, even when the composite fires.
R24 Unfinished night at sync time	If a tester unplugs their phone before their sleep has been finalised, for example unplugging it early and dozing again, the wake trigger syncs a partial night. Sleep duration looks short, the score drops, and the push may tell them to ease off because of data that is simply incomplete.	Before scoring, check that the night's sleep has ended (the last asleep sample is followed by waking); if not, show "sleep still in progress" on the card and send no push; the next run, or a run by hand after waking, completes the night and sends the nudge then.
R25 Password sign-in on a home-screen app	With no email in the MVP (D30), there are no invitation or reset links, so the Safari landing problem disappears. What remains: each tester receives a temporary password from you, a tester who forgets their password is locked out until you set a new one, and passwords bring the usual risks of weak or reused passwords.	Share temporary passwords in person or by private message, never in the onboarding guide; make changing it the first thing testers do in the app; enforce a minimum length and encourage iCloud Keychain's suggested strong password; ask testers to contact you directly if locked out, and keep a way to set a new temporary password with Supabase's admin tools (a short script if the dashboard does not offer it). Test first sign-in, the password change and a reset done by you on a real iPhone at the app spike.

6. Decisions You Still Need to Make

Decided so far

- **D1** MVP users are you plus two or three testers. Consequences: user scoping, consent and basic security move into the must-haves; tester onboarding is needed; validation improves from one person to three or four.
- **D2** Revised: automated sync through an iPhone Shortcuts automation, for both the history backfill and the daily data. There is no manual export in the MVP. Consequences: no app to build; per-user upload tokens become a must-have (R19); the converter script drops out of the MVP and survives only as a fallback if the backfill spike fails; reliability of the automation (R9) becomes the main risk to a daily nudge; backfill depth is set at one year (D26).

- **D3** Data lives and computes on a hosted website you operate (revised twice: your laptop, then each tester's laptop, now hosted). Consequences: consent and security at full strength (R8); multi-user isolation as a correctness requirement (R13); hosting cost and operations (R14); usage evidence comes straight from server logs.
- **D4** Revised: the nudge is sent as a push notification each morning and shown on the card; tapping the notification opens the card. Consequences: push subscriptions, permission and a sending function become must-haves; the nudge reaches testers without them opening the app (R10 largely mitigated); lock-screen privacy (R23) and push reliability (R22) become risks; the morning nudge, and an 8pm follow-up only on days the nudge asked for a change (D32); on a morning with no sync by 11:30am, one reminder push comes first and the nudge follows if a sync arrives before noon (D34), so such a day can reach three pushes.
- **D5** Revised: readiness score built from sleeping HR, overnight HRV and sleep duration. Sleeping HR replaces Apple's resting HR because Apple's value leaves out sleep and is published once, at the end of the day, so at a morning sync the newest one is yesterday's daytime figure. Consequences: all three score inputs now describe last night; simplest inspectable score; no device dependency in the core signal; sleeping HR is derived by the MVP's own SQL from overnight heart-rate samples, so its correctness rests on the pandas check (R16); Apple's resting HR stays in the sync for the illness composite only.
- **D6** Rule-based score with fixed weights. Consequences: explainable from day one, trivially expressed in SQL and checked against the pandas reference (R16); the same weights for every tester keeps evidence comparable; initial weights and thresholds become a decision (D24); tunable or learned weights are design-for-later through stored parameters.
- **D11** Baseline defaults fixed for everyone: a 28-day window that never includes the day being judged, a flag at 2 scaled MADs (MAD multiplied by 1.4826, so it matches a standard deviation), and at least 21 valid days in the window before any flag. Consequences: simple and comparable across testers; per-metric flags are uncommon, so each one carries weight; values stored as settings, so they can change later without a rewrite.
- **D8** Illness and strain composite, weekly digest and nudge feedback all in. Consequences: R7 wording work is now on the must-do list; ingestion carries a few metrics beyond the score; digest generation approach becomes D22.
- **D22** Weekly digest written from a fixed template for the MVP, with a guarded model-written version as a later feature. Consequences: no language-model dependency or wording risk in the MVP; the digest job should be built so the writer is one swappable function.
- **D23** Digest is an in-app page only. Consequences: no email dependency or contact consent; the digest does not leave the app, although the daily nudge now does (D4); reach is bounded by visits, so the digest is a reason to return rather than a reminder to.
- **D24** Card shows status and top drivers, with the Why today screen one tap away. Consequences: the real card can be built from the start rather than a placeholder; opens of that screen are logged as a trust signal; the starting weights and thresholds still wait for the backfill spike.
- **D7** The causal layer is out of the MVP. The method is presented as design, and the experiment registry and event annotation schemas stay in the architecture so it can be built without rework.
- **D12** Daily sync cadence. Consequences: delivered by the Shortcuts automation (D2, D16) rather than by testers doing anything; the endpoint must be idempotent and incremental; the automation's reliability (R9) is what decides whether daily holds.
- **D16** Daily sync runs from two personal automations on each iPhone: a wake trigger when the phone is unplugged from its charger, and a catch-up trigger when the tester first opens an everyday app they choose. The Shortcut acts only between 4am and noon and stops if today's sync already succeeded, so daytime unplugs do nothing. Consequences: no reliance on the iPhone alarm; the wake trigger usually fires within minutes of getting up, and the catch-up guarantees an unlocked run (R9); the server's unfinished-night check (R24) and one-push-per-day rule make the pair safe; nudge timing varies a little by tester and day; onboarding covers two automations and choosing the catch-up app; overnight charging must be confirmed for each user (A11).
- **D27** The tester interface is a mobile-first, installable home-screen web app on the iPhone. Consequences: card and trend view designed for the phone first; long-lived sessions become a must-have; an app icon and full-screen display with no App Store; it is also what makes web push possible on iOS, which D4 now uses; iOS web-app quirks become a risk (R21).
- **D28** Withdrawn: the morning Shortcut no longer opens the card. Push (D4) delivers the nudge and a tap opens the home-screen app directly. Consequences: one prompt per morning instead of two; the Safari landing problem in R21 disappears; there is no backstop if push fails, so delivery monitoring (R22) matters more.
- **D29** The notification carries status and reason ("Ease off today: HRV well below your usual, sleep short"). Consequences: the most useful message without a tap, and the most visible on the lock screen (R23); consent wording and the iOS preview setting become part of onboarding; the R7 wording rule applies to notification text.

- **D30 Revised:** testers sign in with email and password, using Supabase's built-in login, with public sign-up closed. You create each tester's account in the Supabase dashboard and the MVP sends no email: no invitations, and forgotten passwords are reset by you. Consequences: free, with no Apple Developer Program membership or email provider needed; iCloud Keychain gives Face ID autofill on the iPhone; each tester starts with a temporary password shared privately and changes it in the app, so an in-app change-password screen replaces the email reset flow; a locked-out tester waits for you (R25); Shortcut upload tokens unchanged.
- **D19 Analysis** runs on each upload, triggered by the new data. Consequences: the card and push are as fresh as the sync; analysis must be idempotent and scoped to the affected dates; backfill uploads recompute once at the end and never push; at most one morning nudge per user per day, however many syncs arrive.
- **D25** No self-set milestone. Consequences: with no due date either, nothing time-based holds the MVP boundary; D9 becomes the only definition of done and the main defence against scope creep (R18).
- **D9** Success criteria set as a mix over a four-week tester period: hard gates for Feasibility (sync, push, isolation, correctness), target ranges for Signal and Value, a one-tap daily check-in so Signal can be measured on testers, a pre-test check of your backfilled year against objectively disrupted days rather than memory, and a short conversation with each tester at the end. Consequences: the check-in becomes a must-have; your backfill also fetches workout summaries and the event table is used from the start; the targets in Section 1 are confirmed and must not move once results arrive; with four people, Signal and Value show direction rather than proof.
- **D31 Revised:** a two-layer demo. The main demo leads with a story for everyone; the evidence sits in an appendix for those who want it. Main layer: one week from your own data where the tool flagged something before you felt it (a trip, an illness or a hard training block), shown from the user's side, opening with Apple Health's plain daily numbers and then your card for the same day; a few plain results (for example, sync days and week-four retention); testers' own words from the end-of-test conversations; limitations stated briefly in plain language. Appendix: the tap-away arithmetic, trend views with baseline bands, the stage-by-stage SQL-versus-pandas comparison, the disrupted-day check, every gate and target with its result, the limitations in full, and the causal layer as a worked method with its experiment registry (D7). Consequences: polish goes to both the product's feel (card design, notification wording, onboarding) and the appendix evidence; the weekly digest gains prominence as the most readable output; the outputs of every spike are kept, since they become the appendix.
- **D32 Revised:** follow-through is asked on days the nudge asked for a change, from 8pm local time only. A push asks at 8pm and the card shows the question from then; before 8pm the card shows the nudge without it. If still unanswered, the next morning's card asks about yesterday before today's status. Consequences: the one-push-per-day rule becomes one morning nudge plus this follow-up, sent on every change day; the evening push carries no health detail (R23); the server needs each tester's local time; answers arrive the same evening, so the Value evidence stays fresh and complete; a tester who trains in the morning waits until 8pm to answer; train-as-planned days are never asked.
- **D33** The daily check-in can be skipped, and an answer can be changed after it is tapped. Consequences: the card is never blocked by the question; skipped days leave a small prompt on the card for a later answer; changes are allowed until the next morning's check-in; every answer is logged with its time and whether the status had been seen, and the Signal check uses the first answer given before the status wherever possible; fewer complete check-ins are likely, so completion rate is reported with the Signal result.
- **D34 Revised:** if no sync has succeeded by 11:30am local time, a single reminder push asks the tester to run the Shortcut by hand before noon, and a sync before noon still produces today's nudge. Consequences: a missed morning is recoverable the same day; the late nudge is marked as late on the card; the 8pm follow-up still applies on change days; a day gets no nudge only if nothing syncs by noon, when the Shortcut's window closes; a day that needed a reminder can reach three pushes (reminder, late nudge, 8pm follow-up); reminders on consecutive days are flagged to you as a likely broken automation.
- **D35** A missing reading is scored by moving its weight to the other two in proportion to their weights, with the ease off and rest lines unchanged, and the card labelled "Based on 2 of 3 readings". With two or more readings missing, or no sleep recorded, the day gets no status and no push ("Not enough data last night"). A missing night is left out of that reading's 28-night baseline, where the 21-night minimum already applies. Consequences: a night without an HRV reading, which is likely to be common because Apple measures HRV only every few hours, still gets a status; with one reading gone, a single remaining reading can swing the score further, so if the backfill shows one reading alone pushing days to rest, the status is capped at ease off whenever a reading is missing (settled with the D24 numbers); the backfill spike reports how often each reading is missing and reruns the disrupted-day match and the quiet-by-default rate under this rule; partial-data days are reported separately in the Signal results.

- **D36** A reading whose baseline is still building (fewer than 21 valid nights in its 28-night window, D11) is treated as missing under D35. With two or more readings building, the card shows a separate "Learning your normal" state (for example "14 of 21 nights"), with last night's values in plain words but no verdicts, no status, no morning push and no 8pm follow-up; Why today shows the values and charts without the shaded range. The status returns by itself once the window reaches 21 nights, and a day with too little baseline is never shown as Ready. Consequences: with a year of backfill the state is rare, appearing mainly after a gap, a failed backfill or a change of watch; the D9 push gate counts only synced days that have a status, and the tester Signal check leaves out check-ins from days with no status; if HRV is missing on more than 7 of any 28 nights, its baseline never reaches 21, so the backfill spike checks this and, if needed, HRV gets a lower minimum (for example 14 nights) or a longer window (for example 42 nights), settled with the D24 numbers.
- **D37** Supabase stays on the free plan while building and through the four-week test, and moves to Pro after the test. Consequences: the build and the test cost nothing; the free plan has no automatic backups, so you make your own with the Supabase CLI's db dump command to an encrypted folder on your laptop at least weekly and before every database change, and practise one restore before the test; the project can pause after a week of low activity, which daily syncs normally prevent; the 30-day idle sign-out needs Pro, so during the test a session lasts until the tester signs out or uses sign out everywhere; daily backups and the idle sign-out start after the move to Pro.
- **D38** One Supabase project for everything, with a local copy of Supabase (run by the Supabase CLI, which needs Docker) on your laptop and in the automated checks for building and testing database changes. Consequences: no second project to manage; Vercel previews talk to the live database, so every database change is tested on the local copy first, applied only at release and in a backward-compatible way, previews are checked signed in as yourself or a test account, and a backup is taken before every change during the test (R12).
- **D39** On a laptop or desktop browser, the app shows the same single column as on the iPhone, centred and at most 480 points wide, with no separate desktop design. Consequences: nothing extra to design, build or test; the app can be shown from a laptop in the demo; notifications stay iPhone-only.
- **D26** Backfill the last year for every user, in monthly chunks. Consequences: baselines, thresholds and the false-positive check see a full seasonal cycle; the backfill is heavier, so chunks must be resumable (R20); deeper history remains available later by extending the backfill, with nothing redone.
- **D18** The Shortcut posts JSON carrying raw samples rather than daily summaries, with heart rate limited to a fixed 6pm-to-noon window that the server trims to the sleep window. Consequences: all cleaning and derivation live in SQL, so changing a rule is a recompute and every tester is processed identically; the backfill is heavier than summaries would be (R20); each post carries a schema version, backfill flag, month ID and timezone offsets; daytime heart rate can be added later by widening the window.
- **D13** Superseded. Testers now reach their card by logging in to the website. The local-copy approach was chosen and then reversed in favour of a hosted site so a user can log in and see all their data from anywhere.
- **D14** Withdrawn: a website runs in any browser, so platform support is no longer a decision. Testers' iPhones still need to run Shortcuts automations for the daily path.
- **D15** Withdrawn: no desktop packaging for a website. The packaging and Gatekeeper risks (old R13, R14) retire and are replaced by data isolation and hosting risks.
- **D10** A React app on Supabase with cleaning and analysis in SQL, kept simple by three rules: Watch only as the source, the sleep window taken from the Watch's own sleep stages, and sleeping HR defined as the median heart rate during asleep periods. Any step that still proves awkward in SQL moves to a Supabase Edge Function (TypeScript, in the same project), not to a separate service. Reason: one platform to operate. The only local Python is the pandas reference used to check the SQL (R16). Consequences: login, database and row-level security come from Supabase; everything runs on each upload inside Supabase; the pandas check covers cleaning and statistics; Edge Functions have time limits, so backfill is processed one monthly chunk at a time; the code is generated with Claude Code (D20) and the front end is hosted on Vercel (D21).
- **D17** Revised by D20 and settled by D21: Supabase hosts the database, auth, storage and functions; Vercel hosts the front end.
- **D20** Claude Code as the builder. Consequences: the app is generated into a repository you own, so lock-in largely disappears (R17 revised); there is no bundled hosting, so front-end hosting becomes D21; the working style is a terminal, git and deploys with Claude Code doing the writing, which is more hands-on than a hosted builder and worth confirming you are comfortable with.
- **D21** The front end is hosted on Vercel, deployed from your repository with the official Supabase integration. Consequences: zero-config deploys and synced keys; a preview address for every change, which becomes part of deployment discipline (R12); no health data passes through the host; free plan limited to personal, non-commercial use; Vercel's own analytics stays off.

Reversal and its consequences

The tool moved from a local app on each tester's laptop to a hosted website with login. This is a clean resolution of the earlier laptop-versus-card-only conflict and it matches the product brief, but it reverses the privacy benefit of local-first: other people's health data is now on a server you run, so consent and security return as must-haves at full strength (R8), multi-user isolation becomes a correctness requirement (R13), and hosting adds cost and operational load (R14). In exchange you get direct usage evidence without returned files, one version for everyone, and a demo that works from any browser.

Confirmed: score and composite inputs

D5 and D8 pulled slightly against each other, since a composite built from the same three score metrics would add little. Confirmed resolution: the readiness score uses sleeping HR, HRV and sleep duration only (D5, revised); the illness and strain composite also reads Apple's resting HR (yesterday's daytime value) and respiratory rate where available; wrist temperature is excluded from the MVP entirely. Each day therefore carries up to five values, and the Shortcut needs to produce them.

Open

All scoping decisions are now made, and the D9 targets are confirmed. The D24 numbers wait for the backfill spike; the spikes in Section 8 confirm the technical decisions.

Decision	Options	What it changes
D1 Who is the MVP user? (DECIDED)	You plus two or three testers.	User scoping, consent and security become must-haves; onboarding is needed; validation is more credible.
D2 Sync path for the MVP (DECIDED, revised)	Automated sync via an iPhone Shortcuts automation, for both the history backfill and the daily data. No manual export.	No app to build and no third-party dependency; upload tokens become a must-have (R19); converter script only as a fallback; freshness depends on the automation's reliability (R9).
D3 Where data lives and computes (DECIDED, revised twice)	A hosted website with login; data and computation on a server you operate.	Security and consent return as must-haves (R8); multi-user isolation is a correctness requirement (R13); hosting cost and ops (R14); direct evidence via server logs.
D4 Nudge delivery channel (DECIDED, revised)	Push notification each morning plus the card; tapping the notification opens the card.	Push must-haves; analysis on upload (D19); R22 and R23; notification wording D29; replaces D28.
D5 Metric set in the MVP score (DECIDED)	Sleeping HR, overnight HRV, sleep duration (revised: sleeping HR replaces Apple's resting HR, which now feeds the composite only).	No device dependency in the score; extra metrics ingested only for the composite (see flag above).
D6 Score logic (DECIDED)	Rule-based, fixed weights, two thresholds.	Explainable, SQL-friendly, comparable across testers; opens D24 for the actual numbers.
D24 Score numbers and card display (display DECIDED, numbers after the spike)	Card shows status, top drivers and the nudge; full arithmetic one tap away. Starting weights (for example HRV 40%, sleeping HR 35%, sleep 25%) and two thresholds to be set after the backfill spike and frozen before testers start.	Matches the guided-by-default principle; Claude Code can build the real card from the start; the numbers are tuned against the disrupted-day evidence from your backfilled year (A2) and then stay fixed so tester evidence is comparable.
D25 Self-set MVP milestone (DECIDED)	No milestone.	Nothing time-based holds the boundary, so D9 becomes the only definition of done and the main defence against scope creep (R18).

Decision	Options	What it changes
D26 Backfill depth through Shortcuts (DECIDED)	Last year, fetched in twelve monthly chunks, for every user.	Covers every season and the year's trips, illnesses and training blocks, so thresholds (D24) and false positives (R2) are tested across a full cycle; gives the causal layer real events later. About four times the data of 90 days, so resumable chunks matter (R20). Extendable later by running earlier months.
D27 Tester interface (DECIDED)	Mobile-first website installable to the iPhone home screen as a web app.	Phone-first layout for the card and charts; long-lived sessions; enables web push, which D4 now uses; iOS web-app quirks (R21).
D28 Open the card after each sync (WITHDRAWN)	No longer required: push delivers the nudge (D4).	One morning prompt instead of two; Safari landing issue (R21) removed; no backstop if push fails (R22).
D29 What the push notification says (DECIDED)	Status and reason, for example "Ease off today: HRV well below your usual, sleep short".	Most useful without a tap and most visible on the lock screen (R23); consent wording and the iOS preview setting cover it; R7 wording applies.
D30 How testers sign in (DECIDED, revised)	Email and password (Supabase built-in); you create each account and reset passwords yourself, with no email; iCloud Keychain and Face ID autofill on the iPhone.	Free; no developer membership or email provider; an in-app password change replaces email resets; locked-out testers wait for you (R25); Shortcut upload tokens unchanged.
D7 Causal layer in the MVP? (DECIDED)	Out of the MVP. The method is presented as design; schemas stay in the architecture.	Removes the largest data and timeline demand from the MVP. The capstone demonstrates causal reasoning through the design rather than a live result.
D8 Borderline items (DECIDED)	Illness composite, weekly digest and feedback button all in.	R7 wording becomes must-do; composite inputs confirmed (Apple's resting HR and respiratory rate); digest written from a template (D22).
D22 How the weekly digest is written (DECIDED)	Fixed template for the MVP; model-written version as a later feature.	Zero wording risk now (R7); the digest job reads only the insights table, so swapping the writer later touches one function.
D23 Weekly digest delivery (DECIDED)	In-app page only.	No email service or contact consent in the MVP; the digest stays in the app even though the daily nudge is pushed (D4); reach depends on visits, which are already measured (R10).
D9 Success criteria (DECIDED)	A mix: hard gates for Feasibility; target ranges, a daily check-in and end-of-test conversations for Signal and Value; a four-week tester period. Proposed numbers in Section 1.	Defines when the MVP is done (D25); adds the daily check-in as a must-have; targets confirmed and fixed for the test.
D31 Demo emphasis (DECIDED, revised)	Two layers: a story for everyone in the main demo, with the evidence in an appendix.	Polish goes to both the product's feel and the appendix evidence; digest gains prominence; spike outputs become the appendix; limitations brief in the main layer, in full in the appendix.
D32 When follow-through is asked (DECIDED)	On change days only, from 8pm: a push asks and the card shows the question; the next morning's card asks if still unanswered.	Two pushes on every change day (morning nudge and 8pm follow-up); no health detail in the evening push (R23); needs each tester's local time; fresh Value evidence.
D33 Check-in skip and change (DECIDED)	Skippable; changeable until the next morning's check-in.	Card never blocked; answers logged with time and whether status was seen; Signal uses the first pre-status answer; completion rate reported with Signal.

Decision	Options	What it changes
D34 Missed morning sync (DECIDED, revised)	No successful sync by 11:30am local time: one reminder push; a sync before noon still produces a late nudge.	Missed mornings recoverable the same day; no nudge only if nothing syncs by noon; up to three pushes on a reminder day; consecutive reminders flagged to you.
D35 Missing reading (DECIDED)	One reading missing: its weight moves to the other two in proportion, thresholds unchanged, card says "Based on 2 of 3 readings". Two or more missing, or no sleep: no status and no push.	Nights without HRV still get a status; one reading can swing the score further, with an optional cap at ease off settled with D24; partial-data days reported separately.
D36 Baseline still building (DECIDED)	One reading building: treated as missing (D35). Two or more: a "Learning your normal" state with values but no status, push or follow-up; never shown as Ready.	Rare with a year of backfill; the push gate and tester Signal check count only days with a status; HRV may need a lower minimum or a longer window if it is often missing.
D37 Supabase plan (DECIDED)	Free while building and through the test; Pro after the test.	No cost until after the test; own db dump backups weekly and before every change; no idle sign-out until Pro.
D38 Environments (DECIDED)	One Supabase project, plus a local copy for building and testing database changes.	Previews talk to the live database, so changes are tested locally first, applied at release and backward-compatibly, with a backup first.
D39 Laptop layout (DECIDED)	The same single column as the iPhone, centred, at most 480 points wide.	No separate desktop design to build or test; usable for the demo; notifications iPhone-only.
D10 Web stack (DECIDED)	A React app on Supabase (auth, Postgres, row-level security, database functions triggered on upload), generated with Claude Code (D20); cleaning and analysis in SQL, simplified by Watch-only sources, Watch sleep stages as the sleep window, and sleeping HR as the median asleep heart rate; Supabase Edge Functions as the fallback for any awkward step.	One platform, no extra service; simplifying rules keep the SQL tractable, with a pandas reference check (R16); Edge Functions absorb anything SQL handles badly; front end on Vercel (D21).
D10 (record of options considered)	A. React on Supabase, analysis and cleaning in SQL (chosen, with simplifying rules and Supabase Edge Functions as the fallback). B. The same app plus a small Python analysis service (pandas) triggered on upload; easier to verify but a second service to host and monitor. C. Django as a single Python app; pandas built in, but isolation enforced in your own code (R13) and more work for the home-screen app and push. Also considered: Python functions on Vercel, rejected because health data would then pass through Vercel and free-plan limits could break backfill chunks. A no-code platform was dropped once Claude Code became the builder; Python inside the database is not available on Supabase.	A keeps one platform and built-in isolation; the simplifying rules remove most of what made the SQL risky, and Edge Functions cover the rest without a new service. B remains the natural upgrade when later modules (seasonality, the causal layer) outgrow SQL.
D11 Baseline defaults (DECIDED)	Fixed for everyone: a 28-day window (excluding the day judged), a 2 scaled-MAD threshold (MAD multiplied by 1.4826), and at least 21 valid days in the window before a flag is allowed.	Simple, identical for every tester and easy to check in SQL. Scaled MAD keeps per-metric flags uncommon (roughly 1 day in 45 per metric, one direction), which supports quiet-by-default (R2). The 21-day minimum tolerates a week of missed nights.
D12 Sync cadence (DECIDED)	Daily.	Delivered by the two morning Shortcuts automations (D2, D16); holds if the catch-up trigger reliably covers locked-phone failures (A7, R9).

Decision	Options	What it changes
D13 How testers reach their card (DECIDED, revised)	They log in to the website.	Requires accounts, isolation and hosting; removes packaging, tunnels and returned evidence files.
D14 Platforms to support (WITHDRAWN)	Not applicable: any browser.	Only the iPhone Shortcuts path remains platform-specific.
D15 Packaging approach (WITHDRAWN)	Not applicable for a website.	Replaced by deployment discipline (Section 4) and hosting choice (D17).
D17 Hosting platform (REVISED by D20)	Supabase for database, auth, storage and functions; Vercel for the front end (D21).	No builder to bundle hosting now that Claude Code is the tool. Encryption at rest is on every Supabase plan; the free plan runs through the test with your own backups, and Pro follows after the test (D37). One project, with a local copy for development (D38). Region: US East.
D18 Data format (DECIDED)	JSON with raw samples: every reading with timestamp, timezone offset, source and unit, plus schema version, backfill flag and month ID. Heart rate only for a fixed 6pm-to-noon window, trimmed on the server.	All logic stays in SQL, so rule changes are recomputes and testers stay comparable; supports the pandas check (R16). Heavier backfill than summaries (R20); narrowing the heart-rate window is the first fallback.
D19 When the analysis runs (DECIDED)	On each upload, triggered by the new data. The weekly digest stays scheduled.	Freshest card and push; analysis must be idempotent and date-scoped; backfill recomputes once and never pushes; one morning nudge per user per day.
D20 Which AI app builder (DECIDED)	Claude Code generating the app into a repository you own.	Not a hosted builder: you get the code, not a platform. Lock-in (R17) largely disappears; hosting for the front end reopens as D21; you run the terminal, git and deploys yourself, with Claude Code doing the writing.
D21 Front-end hosting (DECIDED)	Vercel, deployed from your repository, with the official Supabase integration.	Zero-config React deploys; Supabase keys kept in sync; a preview address for every change, so you can check it on your phone before testers see it (R12). Free plan is personal, non-commercial use; keep Vercel's analytics off. No health data passes through it.
D16 Daily sync path and triggers (DECIDED, revised)	iPhone Shortcuts automation posting to an authenticated upload endpoint, run by two triggers: the phone being unplugged from its charger, and a catch-up when the tester opens an everyday app of their choice. Acts only between 4am and noon; stops if already synced.	Syncs within minutes of getting up without an alarm; the catch-up guarantees an unlocked run (R9); unfinished nights are caught by the server (R24); timing varies slightly; needs overnight charging (A11).

7. Clarifying Questions and Answers

Answers received on 29 September 2026 are recorded under each question. None remain open.

- What is the class timeline: when is the MVP due, when is the final demo, and how many hours a week can you realistically give it alongside work? **Answer:** No fixed due date for the MVP or the final demo; about 40 hours a week available; no self-set milestone (D25). Time is not the constraint, scope is (R18).
- How much Apple Health history do you have, and do you and your testers wear the watch overnight? (Sleeping HR and respiratory rate depend on it.) **Answer:** Five years of history. You and all three testers wear the watch consistently overnight.
- Do you wear the watch overnight consistently? If not, the overnight metrics will be patchy and the score design changes. **Answer:** Yes, consistently (see question 2).

4. Who are the testers, what watch models do they wear, and do they sleep with the watch on? (Their answers set the metric set, D5.) **Answer:** Three testers, all sleeping with the watch on, wearing Apple Watch Ultra (1st generation) and Apple Watch SE (2nd generation). All five MVP metrics are recorded on both models (A8).
5. Are your testers comfortable with their health data being stored on a server you operate, given the security measures in Section 2, and will they sign a short consent? **Answer:** Yes to both. This settles consent; the security obligations in Section 2 still apply in full (R8).
6. Have you used Claude Code before, and are you comfortable with the working style it implies: a terminal, a git repository, and deploying from it? If not, a half-day getting a hello-world app deployed is the right first investment. **Answer:** Yes, comfortable building in Claude Code.
7. Do your testers use iPhones that can run Shortcuts automations, and will they realistically log in to the site every morning? **Answer:** Yes, all testers have iPhones and can run automations. Daily login is a behaviour to observe rather than assume (R10).
8. How will you know the MVP worked? Are there target numbers you would accept as success? **Decided:** A mix of hard gates and target ranges over a four-week tester period, with a daily check-in and end-of-test conversations (D9, Section 1).
9. For the causal element: are there past events in your own history (a trip, a deload week, a schedule change) that could serve as a retrospective natural experiment, so the method can be demonstrated without waiting for new data? **No longer needed for the MVP:** the causal layer is out (D7). Five years of history is very likely to contain usable events, which makes a retrospective demonstration a strong candidate for the first post-MVP feature.
10. How much does the MVP need to land with a non-technical audience versus demonstrate analytical rigour? The two pull in different directions on where build time goes. **Answer:** A two-layer demo (D31): a story for everyone in the main demo, with the evidence and method in an appendix.

8. Suggested Sequence (investigation, not building)

Every scoping decision is now made and the D9 targets are confirmed, so success cannot be redefined after seeing results. The demo has two layers (D31), a story first and the evidence in an appendix, so keep the outputs of each spike: they become the appendix as well as serving as build checks. Then run four spikes before product code: have Claude Code generate a login-protected two-table React and Supabase app, installable to the iPhone home screen, deploy it to Vercel with the Supabase integration (D21), test first sign-in inside the home-screen app with a temporary password, changing it in the app with Keychain autofill, and a reset done by you (R25), send a test push notification and confirm that tapping it opens the home-screen app still signed in, and verify that a second test account cannot read the first account's rows (R13, R21, R22); the Shortcuts daily sync with a per-user token, run on your phone and one tester's, testing both triggers (how often the unplug run is blocked by the lock and the catch-up takes over, and that a daytime unplug does nothing, and how an unfinished night is handled), with each upload triggering analysis and a single morning push, and the 8pm follow-up arriving only on change days (A7, R9, R19, R24, D16, D19); the one-year Shortcuts backfill in monthly chunks, timed on your phone and a tester's, including a deliberately interrupted run to prove it resumes (D26, R20), and confirming that backfill never sends a push; and the SQL baseline over your backfilled history, checked against the pandas reference (R16) and against the disrupted-day evidence from your travel, calendar and workouts (A2, R2). At 40 hours a week the spikes are days of work, not weeks, and their results are the last input needed to fix the MVP boundary.